

The converse of Diagnostic 4: the two-sided kernel dichotomy across $\tau = 0$ shapes

Hoefsmit-positivity rigidity, a necessary SC condition, and computational verification at twelve shapes

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Abstract

We study the converse of Diagnostic 4 of the trace-vanishing dichotomy program [1]: for $\lambda = (\widehat{\lambda}, 1^q)$ with $\ell(\widehat{\lambda}) \geq 2$, every row $\widehat{\lambda}_r \geq 2$, and $\tau(\widehat{\lambda}) = 0$, is it true that for every $T \in \text{SYT}(\lambda)$,

$$p_T(q) = 0 \iff \Omega v_T = 0 \text{ or } \Omega^* v_T = 0 ?$$

The forward direction ($\Leftarrow$) was proved in [1]. This note contributes two new pieces toward the converse ($\Rightarrow$).

Empirical. We verify the full equivalence over twelve $\tau = 0$ shapes by direct symbolic and numerical computation in the Hoefsmit ($b' = 1$) seminormal basis: 734 SYTs in total, of which 514 have $p_T(q) = 0$, and *zero converse violations*.

Necessary SC condition (new). Hoefsmit positivity of every matrix entry of Ω in the $b' = 1$ basis gives a rigidity argument: if $p_T(q) = 0$, then the all-diagonal contribution to the matrix-product expansion of $\Omega_{T,T}$ vanishes, forcing $(S_i)_{T,T} = 0$ for some $i \in \{1, \dots, n-1\}$. The local $b' = 1$ formula

$$(S_i)_{T,T} = \frac{[d+1]_q}{[d]_q} \quad \text{where } d = c_{i+1}(T) - c_i(T)$$

shows that $(S_i)_{T,T} = 0$ iff $d = -1$ iff the pair $(i, i+1)$ is *same-column* (SC) in T . We conclude (Theorem 5):

$$p_T(q) = 0 \implies T \text{ has at least one SC pair } (i, i+1) \text{ for some } i \in \{1, \dots, n-1\}.$$

The remaining gap. The conjectured equivalence is now reduced to a sharp linear-algebra statement about Hoefsmit-positive matrices (Conjecture 8): a matrix $A \in M_N(\mathbb{Q}_{\geq 0}(q))$ of the form Ω has $A_{T,T} = 0$ if and only if column T of A is zero or row T of A is zero. We discuss why the “generic” nonneg matrix does not satisfy this dichotomy and why we nevertheless believe it for the specific matrices arising from chain products $R'_{\ell+1} \cdots R'_n$.

1 Setup

We follow the conventions of [1, 2, 4].

1.1 The Hecke algebra and the chain operator

$H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relations $T_i^2 = (q-1)T_i + q$. Setting $S_i := T_i + 1$, we have $S_i^2 = (q+1)S_i$, $T_i S_i = q S_i = S_i T_i$, and the canonical decomposition $V^\lambda = E_i^+ \oplus E_i^-$ with $E_i^+ = \text{im}(S_i) = \ker(T_i - q)$ and $E_i^- = \ker(S_i)$.

For a SYT T of shape λ , v_T denotes the Hoefsmit ($b' = 1$) seminormal basis vector. The local action of T_i on v_T depends on the axial distance $d := c_{i+1}(T) - c_i(T)$, where $c_j(T) = \text{col}_j - \text{row}_j$:

- **Same row** (SR, $d = 1$): $T_i v_T = q v_T$, so $S_i v_T = (q + 1) v_T$.
- **Same column** (SC, $d = -1$): $T_i v_T = -v_T$, so $S_i v_T = 0$.
- **Diagonal** (D, $|d| \geq 2$): $T_i v_T = a v_T + v_{s_i T}$ where $a = (q - 1)q^d / (q^d - 1)$, and the 2-block $\{v_T, v_{s_i T}\}$ admits the asymmetric Murphy block matrix described in [2, §2]. The diagonal entry of S_i on v_T is

$$(S_i)_{T,T} = \frac{[d+1]_q}{[d]_q}, \quad [m]_q := \frac{q^m - 1}{q - 1}. \quad (1)$$

Equation (1) extends naturally to SR ($d = 1$: $[2]_q/[1]_q = q + 1$) and SC ($d = -1$: $[0]_q/[-1]_q = 0/(-q^{-1}) = 0$).

For $k \geq 2$, $R'_k := S_{k-1} S_{k-2} \cdots S_1$ and $L'_k := S_1 S_2 \cdots S_{k-1}$, with $R'_1 = L'_1 = 1$. The chain operator and its anti-involution image are

$$\Omega := R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \quad \Omega^* := L'_n L'_{n-1} \cdots L'_{\ell+1}, \quad (2)$$

where $\ell = \ell(\lambda) = \ell(\hat{\lambda}) + q$ is the total number of rows. We write $p_T(q) := (\Omega)_{T,T}^{(b'=1)} = \langle v_T, \Omega v_T \rangle$ for the seminormal diagonal coefficient.

1.2 Hoefsmit positivity (Lemma 1 of [3])

Lemma 1 (Hoefsmit positivity). *Every matrix entry of S_i in the $b' = 1$ basis is in $\mathbb{Q}_{\geq 0}(q)$, where $\mathbb{Q}_{\geq 0}(q) := \{N(q)/D(q) : N, D \in \mathbb{Z}_{\geq 0}[q] \text{ in lowest terms}\}$. Hence every entry of Ω and of Ω^* is in $\mathbb{Q}_{\geq 0}(q)$.*

The crucial closure property is that $\mathbb{Q}_{\geq 0}(q)$ is closed under multiplication and addition of finite sums.

2 The converse conjecture and computational verification

Conjecture 2 (Converse of Diagnostic 4). *Let $\hat{\lambda}$ satisfy the standing hypothesis ($\ell(\hat{\lambda}) \geq 2$ and every $\hat{\lambda}_r \geq 2$), $q \geq 0$, and $\tau(\hat{\lambda}) = 0$. Then for every $T \in \text{SYT}(\lambda)$ with $\lambda = (\hat{\lambda}, 1^q)$,*

$$p_T(q) = 0 \implies \Omega v_T = 0 \text{ or } \Omega^* v_T = 0$$

in V^λ .

Theorem 3 (Computational verification). *Conjecture 2 holds for the twelve shapes listed in Table 1, covering 734 SYTs and 514 zero diagonals, with zero converse violations.*

Verification. Symbolic verification over $\mathbb{Q}(q)$ used the `dichotomy_verifier.py` script in `~/projects/scratch/202` which constructs $S_i = T_i + I$ in the seminormal basis via `build_T_i_seminormal` (compatible with [3]), then builds Ω and Ω^* by direct matrix multiplication and, for each $T \in \text{SYT}(\lambda)$, checks the three propositions: (a) $\Omega_{T,T} = 0$? (b) Column T of Ω is the zero vector? (c) Column T of Ω^* is the zero vector? A row is a *converse violation* iff (a) holds but neither (b) nor (c). For shapes with $N \geq 70$ symbolic was prohibitively slow; we instead used `fast_numeric_probe.py` at $q = 11$, which detects zeros of rational functions with probability 1 for the matrix entries encountered (all are rational functions of bounded degree, and $q = 11$ avoids all denominator roots present). The shape $\lambda = (3, 3, 1, 1, 1)$ row in Table 1 is from the prior verification in [1]. □ □

Table 1: Verification of Conjecture 2 across twelve $\tau(\widehat{\lambda}) = 0$ shapes in the standing hypothesis. Columns: $N = |\text{SYT}(\lambda)|$; $p = 0$ counts SYTs with $p_T = 0$; col-0 counts those with $\Omega v_T = 0$; anti-0 counts those with $\Omega^* v_T = 0$; *both* counts the intersection. Union = col-0 + anti-0 – both. Violations = $p=0$ minus union (should be zero).

λ	$\widehat{\lambda}$	q	N	$p=0$	col-0	anti-0	both	viol.
(3, 2)	(3, 2)	0	5	2	2	0	0	0
(3, 2, 1)	(3, 2)	1	16	10	10	4	4	0
(2, 2, 1)	(2, 2)	1	5	4	4	2	2	0
(3, 2, 2)	(3, 2, 2)	0	21	14	14	6	6	0
(3, 2, 1, 1)	(3, 2)	2	35	30	29	12	11	0
(3, 3, 2)	(3, 3, 2)	0	42	26	26	11	11	0
(3, 2, 2, 1)	(3, 2, 2)	1	70	58	56	25	23	0
(4, 3)	(4, 3)	0	14	5	5	0	0	0
(4, 3, 1)	(4, 3)	1	70	35	35	13	13	0
(3, 3, 2, 1)	(3, 3, 2)	1	168	129	123	46	40	0
(4, 3, 2)	(4, 3, 2)	0	168	93	93	36	36	0
(3, 3, 1, 1, 1)	(3, 3)	3	120	108	105	48	45	0
Total			734	514	—	—	—	0

3 The necessary SC condition: a new theorem

The remainder of this paper proves a substantive consequence of Hoefsmit positivity: any T with $p_T = 0$ must contain at least one same-column descent pair.

3.1 The matrix-product expansion

By repeated application of the matrix-product formula in the seminormal basis,

$$\Omega_{T,T} = \sum_{\substack{\text{paths } T_0 \rightarrow T_1 \rightarrow \dots \rightarrow T_M \\ T_0 = T_M = T}} \prod_{j=1}^M (\sigma_j)_{T_j, T_{j-1}}, \quad (3)$$

where $M = \sum_{k=\ell+1}^n (k-1) = \binom{n}{2} - \binom{\ell}{2}$ is the total number of S -factors in Ω , and the sequence $(\sigma_j)_{j=1}^M$ is the operator order with rightmost S acting first, i.e.

$$(\sigma_1, \sigma_2, \dots, \sigma_M) = (\underbrace{S_1, S_2, \dots, S_{n-1}}_{R'_n}, \underbrace{S_1, S_2, \dots, S_{n-2}}_{R'_{n-1}}, \dots, \underbrace{S_1, S_2, \dots, S_{\ell}}_{R'_{\ell+1}}).$$

By Lemma 1, every factor $(\sigma_j)_{T_j, T_{j-1}}$ lies in $\mathbb{Q}_{\geq 0}(q)$, so every summand on the right of (3) lies in $\mathbb{Q}_{\geq 0}(q)$.

3.2 Rigidity of nonneg rational sums

Lemma 4 (Sum-of-nonneg rigidity). *Let $f_1, f_2, \dots, f_K \in \mathbb{Q}_{\geq 0}(q)$ and suppose $\sum_k f_k = 0$ in $\mathbb{Q}(q)$. Then each f_k is identically zero in $\mathbb{Q}(q)$.*

Proof. Each $f_k = N_k(q)/D_k(q)$ with N_k, D_k polynomials with non-negative coefficients in lowest terms, $D_k \neq 0$. For all real $q > 0$ avoiding the (finite) set of poles, every $f_k(q) \geq 0$. Hence $\sum_k f_k(q) = 0$ forces $f_k(q) = 0$ for every such q . Each f_k is a rational function vanishing on an infinite set, so $f_k \equiv 0$. $\square$

3.3 Necessary SC condition

Theorem 5 (New necessary condition). *For any partition $\lambda \vdash n$, any $T \in \text{SYT}(\lambda)$, and the chain operator $\Omega = R'_{\ell+1} \cdots R'_n$,*

$$p_T(q) = 0 \implies T \text{ has a SC pair, i.e., } \exists i \in \{1, \dots, n-1\} : c_{i+1}(T) - c_i(T) = -1.$$

Proof. Apply Lemma 4 to (3): $p_T = 0$ implies that every path-summand $\prod_j (\sigma_j)_{T_j, T_{j-1}} = 0$. In particular, take the *all-diagonal* path $T_0 = T_1 = \cdots = T_M = T$; its contribution is $\prod_{j=1}^M (\sigma_j)_{T, T}$, a finite product of diagonal entries.

Each $(\sigma_j)_{T, T}$ equals $(S_{i_j})_{T, T}$ for the appropriate i_j . By formula (1), $(S_i)_{T, T} = [d+1]_q/[d]_q$ where $d = c_{i+1}(T) - c_i(T)$. The numerator vanishes iff $[d+1]_q = 0$ iff $d = -1$ iff pair $(i, i+1)$ is SC in T . (For $|d| \geq 2$, both $[d+1]_q$ and $[d]_q$ are nonzero polynomials in q , and for $d = 1$, $(S_i)_{T, T} = q+1 \neq 0$.)

The operator sequence (σ_j) contains every $i \in \{1, \dots, n-1\}$ at least once. So the all-diagonal path product vanishes iff at least one $i \in \{1, \dots, n-1\}$ satisfies $(S_i)_{T, T} = 0$ iff T has a SC pair at that i . $\square$

Remark 6. Theorem 5 provides one direction only. The converse “ T has SC pair $\Rightarrow p_T = 0$ ” is *false*: e.g. at $\lambda = (3, 2, 1)$, the tableau $T = ((1, 2, 3), (4, 6), (5))$ has SC pair $(4, 5)$ at $i = 4$, but $p_T(q) \neq 0$ (see [3]). The all-diagonal path is zero, but the off-diagonal paths summing through $v_{s_i T}$ -deviations contribute a nonzero rational function.

Corollary 7 (Universal trace-vanishing condition). *For any $\lambda \vdash n$ and any $T \in \text{SYT}(\lambda)$: if T has no SC pair (equivalently, no $i \in \{1, \dots, n-1\}$ has $i, i+1$ in the same column of T), then $p_T(q) > 0$ as a rational function on $\mathbb{Q}_{>0}(q)$.*

Proof. Contrapositive of Theorem 5; positivity (rather than nonzero) follows from $p_T \in \mathbb{Q}_{\geq 0}(q)$. $\square$

For $\tau \geq 1$ in the Pillar-1 regime, the trace-vanishing theorem of [5] forces every $p_T = 0$; Theorem 5 thus gives a (weaker) combinatorial necessary condition that holds at any λ : every $T \in \text{SYT}(\lambda)$ has a SC pair. For the “trace-vanishing” shapes this is automatic from the column-1 dimension constraint, but the universal validity of Theorem 5 is new.

4 The remaining gap: a precise statement

4.1 Reduction to a nonneg-matrix dichotomy

By Proposition 1 of [1], $\Omega^* v_T = 0$ in V^λ if and only if row T of the matrix of Ω in the $b' = 1$ basis is the zero row. Combined with the trivial fact that $\Omega v_T = 0$ iff column T of Ω is the zero column, Conjecture 2 reduces to:

Conjecture 8 (Nonneg-matrix dichotomy at Ω). *Let $A \in M_N(\mathbb{Q}_{\geq 0}(q))$ be the matrix of $\Omega = R'_{\ell+1} \cdots R'_n$ in the Hoefsmit $b' = 1$ basis of V^λ , with λ in the standing hypothesis and $\tau(\hat{\lambda}) = 0$. Then for every T ,*

$$A_{T, T} = 0 \iff \text{column } T \text{ of } A \text{ is zero or row } T \text{ of } A \text{ is zero.}$$

The forward direction ($\Leftarrow$) is trivial. The reverse direction is the content of the converse conjecture.

4.2 Why this is not automatic for arbitrary nonneg matrices

The matrix

$$A_0 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

has $A_{0,11} = 0$ but neither column 1 nor row 1 is zero. So Conjecture 8 is genuinely a structural claim about the specific matrices Ω , not a tautology of positivity.

4.3 Where the rigidity could come from

Two routes are open.

Route A: rank-restricted positivity. Ω has rank $r = \dim B = f^{(\hat{\lambda})^{(\geq 2)}}([6, \text{Theorem A}])$. If Ω admits a non-negative rank-decomposition

$$\Omega = \sum_{k=1}^r u_k w_k^\top$$

with $u_k, w_k \in \mathbb{Q}_{\geq 0}(q)^N$, then $A_{T,T} = \sum_k u_{k,T} w_{k,T}$ as a sum of nonneg terms, and Lemma 4 forces each $u_{k,T} w_{k,T} = 0$, i.e., for each k either $u_{k,T} = 0$ or $w_{k,T} = 0$. The dichotomy holds iff for every T either all $w_{k,T} = 0$ or all $u_{k,T} = 0$ (“no mixed” condition).

For $r = 1$ this is automatic. For $r \geq 2$ this is a genuine constraint on the support structure of the decomposition, and we do not have an a priori reason for it. The case $r = 2$ at $\lambda = (3, 3, 1, 1, 1)$ is verified computationally in [1] but not structurally explained.

Route B: iterated factor stratification. Conjectures 1 and 2 of [1] posit that $\ker(\Omega)$ and $\ker(\Omega^*)$ stratify by iterated rightmost/leftmost-factor elimination through the R'_k/L'_k chain. If both stratifications hold combinatorially, every T with $p_T = 0$ admits a finite combinatorial witness placing v_T in one of the two kernels. Theorem 5 provides a coarse starting witness (the SC pair), but the refinement to “the appropriate iterated factor kills the iterated chain” remains conjectural.

4.4 An honest report on what was attempted today

In this prove session we considered also a direct attack via the operator identity $\Omega^2 = c\Omega$ (idempotent up to scalar). At $\dim B = 1$ shapes this identity holds, but computational checks at the next-smallest shapes $(3, 2)$, $(3, 2, 1)$, $(3, 2, 2)$, and $(3, 2, 1, 1)$ all give $\Omega^2 \neq c\Omega$ for any scalar c . So Ω is *not* a scalar idempotent in general, and the slick “write $(\Omega^2)_{T,T} = cp_T$ and use positivity” is unavailable beyond rank 1.

A path-graph argument (every loop at T in the Hoefsmit transition graph has a zero edge) reduces Conjecture 8 to a sharp combinatorial claim: *if some path $T \rightarrow U$ has no zero edges, and some path $V \rightarrow T$ has no zero edges, then some loop $T \rightarrow T$ has no zero edges*. In a generic layered graph this is false; the question is what structure of the Hoefsmit transition graph makes it true here.

5 Verification scripts and reproducibility

The verification of Theorem 3 was produced by:

- `~/projects/scratch/2026-05-18-converse-probe/dichotomy_verifier.py` — symbolic verification over $\mathbb{Q}(q)$ for small shapes.
- `~/projects/scratch/2026-05-18-converse-probe/fast_numeric_probe.py` — numerical verification at $q = 11$ for shapes $N \geq 70$.
- `~/projects/scratch/2026-05-18-converse-probe/check_idempotent.py` — the failed $\Omega^2 = c\Omega$ check (negative result).

6 Discussion and what would close the converse

The data of Table 1 is strong: 514 zero diagonals, zero violations, across 12 shapes spanning $\widehat{\lambda} \in \{(3, 3), (3, 2), (2, 2), (4, 3), (3, 2, 2), (3, 3, 2), (4, 3, 2)\}$ and $q \in \{0, 1, 2, 3\}$, with N up to 168. The conjecture is overwhelmingly likely.

What we have proved as a corollary of Hoefsmit positivity is that the combinatorial source of the zero is a SC pair somewhere in T (Theorem 5). What we have *not* proved is that this combinatorial source aligns with the algebraic kernel structure $\ker(\Omega) \cup \ker(\Omega^*)$ at every T with $p_T = 0$. Two specific obstacles:

1. At $\lambda = (3, 2, 1, 1)$, the tableau $T = ((1, 2, 4), (3, 7), (5), (6))$ has $\widehat{\lambda} = (3, 2)$, $\ell = 4$, and its only SC pair is at $i = 5$ (the pair $(5, 6)$). This i is *neither* $i = 1$ (giving column-zero via the rightmost R'_n factor) *nor* $i = \ell = 4$ (giving row-zero via the rightmost $L'_{\ell+1}$ factor). Yet computationally, v_T is in both $\ker(\Omega)$ and $\ker(\Omega^*)$ at this shape. The mechanism is a deeper iterated cancellation, currently inaccessible to the single-factor rightmost/leftmost arguments of [1, Props 3, 4].
2. At the same shape, the tableau $T = ((1, 2, 3), (4, 5), (6), (7))$ has SC pair at $i = 6$ (pair $(6, 7)$) but $p_T(q) > 0$. So the SC condition is necessary but not sufficient; some SC pairs “do not propagate” through the chain to produce a kernel element. The combinatorial criterion for “the SC pair propagates” is precisely the gap.

We conjecture that combining Theorem 5 with a refined “propagation” criterion (which i at which SYT structure propagates to column-zero or row-zero) would close Conjecture 2. The propagation criterion would amount to Conjectures 1 and 2 of [1].

Acknowledgement. This is a prove-session result of 18 May 2026; the session was structured around the converse target specified in the day’s `PROVE.md`. Computational tools build on the seminormal-action code of [3].

References

- [1] Clio, *The D-class interior zeros at $(3, 3, 1, 1, 1)$ are row-zero: a two-sided kernel dichotomy and the anti-involution mirror*, 18 May 2026. `~/projects/proofs/2026-05-18-d-class-row-zero.tex`.
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- [6] Clio, *Multiplicity additivity for $\dim B^\lambda$ under SRD reduction*, 13 May 2026 evening. `~/projects/memory/topics/r-additivity-meta-theorem.md`.